Polinomi di Taylor

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + R_n(x)$$

$$R_n(x) = (x-x_0)^n$$

$$R_n(x) = \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(c) \qquad c \in I(x_0)$$

Calcolare $\sqrt[5]{e}$ con un errore minore di 10^{-4}

$f(x) = e^x \qquad x_0 = 0$

$f(0) = 1 \qquad f'(x) = e^x \qquad f'(0) = 1$

$f''(x) = e^x \qquad f''(0) = 1 \qquad f'''(x) = e^x$

$f^{IV}(x) = e^x \qquad f^{IV}(0) = 1 \qquad f'''(0) = 1$

$$f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{f^{V}(c)}{5!}x^5$$

$c \in (0, 1)$

$$f\left(\frac{1}{5}\right) = \sqrt[5]{e} = 1 + \frac{1}{5} + \frac{1}{50} + \frac{1}{5^3 \cdot 6} + \frac{1}{5^4 \cdot 24} + \frac{e^c}{120} \cdot \frac{1}{5^5}$$

$$\frac{e^c}{(n+1)!}x^{n+1} < 10^{-4} \qquad c \in (0,1)$$

$$\frac{e^c}{(n+1)!} \cdot \frac{1}{5^{n+1}} < 10^{-4}$$

$n=1$ $\quad \frac{e^c}{2!} \cdot \frac{1}{25} < 10^{-4}$ $\quad e^c < 3$, $c \in (0,1)$

$\frac{e^c}{2!} \cdot \frac{1}{25} < \frac{3}{2!} \cdot \frac{1}{25} < 10^{-4}$ $\quad$ no

$\frac{3}{2!} \cdot \frac{1}{25} = 0{,}06$

$n=2$ $\quad \frac{3}{3!} \cdot \frac{1}{125} < 10^{-4}$ $\quad$ no

$\frac{3}{3!} \cdot \frac{1}{125} = 0{,}004$

$n=3$ $\quad \frac{3}{4!} \cdot \frac{1}{5^4} < 10^{-4}$ $\quad$ no

$\frac{3}{4!} \cdot \frac{1}{5^4} = 0{,}0002$

$n=4$ $\quad \frac{3}{5!} \cdot \frac{1}{5^5} < 10^{-4}$ $\quad$ Sí

CALCOLO di LIMITI

$$\lim_{x\to 0} \frac{e^x - 1 - \sin x - x^2/2}{x^2}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + o(x^3)$$

$$\sin x = x - \frac{x^3}{3!} + o(x^4)$$

$$\lim_{x\to 0} \left[1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3) - 1 + \right.$$
$$\left. - \left(x - \frac{x^3}{6} + o(x^4) \right) - \frac{x^2}{2} \right] : x^2$$

$$\lim_{x\to 0} \frac{\frac{x^3}{6} + o(x^3) + \frac{x^3}{6} + o(x^4)}{x^2}$$

$$\lim_{x\to 0} \frac{\frac{x^3}{3} + o(x^3)}{x^2} = 0$$

$$\lim_{x\to 0} \frac{x^2 \cos x + 1 - e^{x^2}}{x^2 \sin^2 x}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + o(x^5)$$

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + o(x^4)$$

$$\lim_{x\to 0} \frac{x^2\left(1-\frac{x^2}{2}+\frac{x^4}{4!}+o(x^5)\right)+1-1-x^2-\frac{x^4}{2}+o(x^4)}{x^2\left(x^2+o(x^2)\right)}$$

$\sin^2 x \sim x^2$

$$\lim_{x\to 0} \frac{x^2-\frac{x^4}{2}+\frac{x^6}{4!}+o(x^7)-x^2-\frac{x^4}{2}+o(x^4)}{x^4+o(x^4)}$$

$$\lim_{x\to 0} \frac{-x^4+o(x^4)}{x^4+o(x^4)} = -1$$

$$\lim_{x\to 0} \frac{-x^4+\frac{x^6}{4!}+o(x^7)+o(x^4)}{x^4+o(x^4)}$$

$$\lim_{x\to 0} \frac{x^4\left(-1+\frac{x^2}{4!}+o(x^3)+o(1)\right)}{x^4+o(x^4)}$$

$$\lim_{x\to 0} \frac{x\sin x+\log(1-x^2)}{x^2(2x+x^2)^2}$$

$$\log(1+x) = x-\frac{x^2}{2}+\frac{x^3}{3}+o(x^3)$$

$$\lim_{x\to 0} \frac{x\left(x-\frac{x^3}{3!}+o(x^4)\right)+(-x^2)-\frac{(-x^2)^2}{2}+o(x^4)}{x^4(2+x)^2}$$

$$\lim_{x\to 0} \frac{x^2-\frac{x^4}{6}+o(x^5)-x^2-\frac{x^4}{2}+o(x^4)}{x^4(2+x)^2}$$

$$\lim_{x\to 0} \frac{-\frac{2}{3}x^4+o(x^4)}{x^4(2+x)^2} = -\frac{1}{6}$$

$$\lim_{x\to 0} \frac{\sin x-\cos x+1}{2x+x^2+1-\frac{e^x-1}{x}}$$

$$\lim_{x\to 0} \frac{x-\frac{x^3}{3!}+o(x^4)-1+\frac{x^2}{2!}+o(x^3)+1}{2x+x^2+1-\frac{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+o(x^3)-1}{x}}$$

$$\lim_{x\to 0} \frac{x-\frac{x^3}{6}+o(x^4)+\frac{x^2}{2}+o(x^3)}{2x+x^2+1-1-\frac{x}{2}-\frac{x^2}{6}+o(x^2)}$$

$$\lim_{x\to 0} \frac{x+\sigma(x)}{\frac{3}{2}x+\frac{5}{6}x^2+\sigma(x^2)} = \frac{2}{3}$$

$$\lim_{x\to 0} \frac{\log(1+x^2)-\sin^2 x}{e^{x^2}-1-x^2}$$

$$e^{x^2}-1 \sim x^2$$

$$\lim_{x\to 0} \frac{x^2-\frac{x^4}{2}+\sigma(x^4)-\left(x-\frac{x^3}{3!}+\sigma(x^4)\right)^2}{1+x^2+\frac{x^4}{2!}+\sigma(x^4)-x^2-1}$$

$$\lim_{x\to 0} \frac{x^2-\frac{x^4}{2}+\sigma(x^4)-x^2+\frac{2x^4}{3!}+\sigma(x^4)}{\frac{x^4}{2}+\sigma(x^4)}$$

$$\lim_{x\to 0} \frac{-\frac{1}{6}x^4+\sigma(x^4)}{\frac{x^4}{2}+\sigma(x^4)} = -\frac{1}{3}$$

$$\lim_{x\to 1} \frac{\cos\left(\frac{\pi}{2}x\right)\sin(x-1)}{\sin^2(\pi x)}$$

$x-1=t \Rightarrow x=t+1$

$$\lim_{t\to 0} \frac{\cos\left(\frac{\pi}{2}(t+1)\right)\sin t}{\sin^2(\pi(t+1))}$$

$$\lim_{t\to 0} \frac{\cos\left(\frac{\pi}{2}t+\frac{\pi}{2}\right)\sin t}{\sin^2(\pi t+\pi)}$$

$$\lim_{t\to 0} \frac{-\sin\frac{\pi}{2}t\cdot\sin t}{(-\sin\pi t)^2}$$

$$\lim_{t\to 0} \frac{-\frac{\pi}{2}t\cdot t+o(t^2)}{\pi^2 t^2+o(t^2)} = -\frac{1}{2\pi}$$

$$\lim_{x\to+\infty} \frac{x^2\left(e^{\frac{1}{x^2}} - \cos\frac{1}{x} - \frac{3}{2x^2}\right)}{\sin^2\frac{1}{x}}$$

$t = \frac{1}{x}$

$$\lim_{t\to 0} \frac{\frac{1}{t^2}\left(e^{t^2} - \cos t - \frac{3}{2}t^2\right)}{\sin^2 t}$$

$$\lim_{t\to 0} \left[\frac{1}{t^2}\left(1 + t^2 + \frac{(t^2)^2}{2!} + o(t^4) - 1 + \frac{t^2}{2} - \frac{t^4}{4!} + o(t^4) - \frac{3}{2}t^2\right)\right] : \left(t^2 + o(t^2)\right)$$

$$\lim_{t\to 0} \frac{\frac{1}{t^2}\left(\frac{t^4}{2} + o(t^4) - \frac{t^4}{24} + o(t^4)\right)}{t^2 + o(t^2)}$$

$$\lim_{t\to 0} \frac{\frac{t^2}{2} - \frac{t^2}{24} + o(t^2)}{t^2 + o(t^2)} = \frac{11}{24}$$

$$\lim_{x\to 0^+} \frac{1}{x^3}\left(\frac{\sin x}{x} - \frac{x}{\sin x}\right)$$

$$\lim_{x\to 0^+} \frac{1}{x^3}\left(\frac{\sin^2 x - x^2}{x\sin x}\right)$$

$$\lim_{x\to 0^+} \frac{1}{x^a}\left(\frac{\left(x-\frac{x^3}{6}+o(x^4)\right)^2-x^2}{x(x+o(x))}\right)$$

$$\lim_{x\to 0^+} \frac{1}{x^a}\left(\frac{x^2-\frac{x^4}{3}+o(x^4)-x^2}{x^2+o(x^2)}\right)$$

$$\lim_{x\to 0^+} \frac{1}{x^a}\left(-\frac{1}{3}x^2+o(x^2)\right)$$

$$\lim_{x\to 0^+} -\frac{1}{3}x^{2-a} = -\begin{cases} 2-a>0 \quad a<2 \to 0 \\ 2-a<0 \quad a>2 \to -\infty \\ a=2 \quad -\frac{1}{3} \end{cases}$$

$$\lim_{x\to 0} \frac{\sqrt{1-x^4} - \cos x^2}{x^2 \sin x - x \log(1+x^2)}$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}x^3 + \dots + \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}x^n + o(x^n)$$

- $(1-x^4)^{1/2} = 1 + \frac{1}{2}(-x^4) + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!}(-x^4)^2 + o(x^8)$
$= 1 - \frac{1}{2}x^4 - \frac{1}{8}x^8 + o(x^8)$

- $\cos x^2 = 1 - \frac{(x^2)^2}{2} + \frac{(x^2)^4}{4!} + o(x^8)$

$$\lim_{x\to 0} \frac{1 - \frac{1}{2}x^4 - \frac{1}{8}x^8 + o(x^8) - \left(1 - \frac{x^4}{2} + \frac{x^8}{24} + o(x^8)\right)}{x^2\left(x - \frac{x^3}{3!} + o(x^4)\right) - x\left(x^2 - \frac{x^4}{2} + o(x^4)\right)}$$

$$\lim_{x\to 0} \frac{-\frac{1}{6}x^8 + o(x^8)}{x^3 - \frac{x^5}{3!} + o(x^6) - x^3 + \frac{x^5}{2} + o(x^5)}$$

$$\lim_{x\to 0} \frac{-\frac{1}{6}x^3 + o(x^3)}{\frac{1}{3}x^3 + o(x^3)} = 0$$

$$\lim_{x\to 0} \frac{e^{\sin x} - e^x}{\sin^3(2x)}$$

$$e^{\sin x} = 1 + \sin x + \frac{\sin^2 x}{2} + \frac{\sin^3 x}{3!} + o(\sin^3 x)$$

$$= 1 + \left(x - \frac{x^3}{3!} + o(x^4)\right) + \frac{1}{2}\left(x - \frac{x^3}{3!} + o(x^4)\right)^2 +$$

$$+ \frac{1}{3!}\left(x - \frac{x^3}{3!} + o(x^4)\right)^3 + o(\sin^3 x)$$

$$= 1 + x - \frac{x^3}{6} + o(x^4) + \frac{1}{2}x^2 - \frac{1}{6}x^4 + o(x^4) +$$

$$+ \frac{1}{6}x^3 + o(x^3) + o(x^3)$$

$$\lim_{x\to 0} \left(1 + x + \frac{1}{2}x^2 - \frac{1}{6}x^4 + o(x^4) - 1 - x - \frac{x^2}{2} + \right.$$

$$\left. - \frac{x^3}{3!} - \frac{x^4}{4!} + o(x^4)\right) : \left(8x^3 + o(x^3)\right)$$

$$\lim_{x\to 0} \frac{-\frac{x^3}{6} + o(x^3)}{8x^3 + o(x^3)} = -\frac{1}{48}$$

$$\lim_{x\to 0} \frac{\sin(\log(1+2x)) - e^{2x} + 1}{\tan(x^2)} = -4$$

$f(x) = \cos x + x^4 \qquad x = 0$

$f'(x) = -\sin x + 4x^3 \qquad f'(0) = 0$

$f(x) - f(x_0) = f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2$

$f(x) - f(0) = \frac{1}{2} f''(0) \cdot x^2$

$f''(x) = -\cos x + 12x^2 \qquad f''(0) = -1$

$f(x) - f(0) = -\frac{1}{2} x^2 < 0$

$\Rightarrow f(x) < f(0) \Rightarrow x = 0$ max relat.

Sia $f(x) = \log(1+2x^2) + 4(\cos x - 1)$

a) scrivere il polinomio di Mac Laurin di f fino al primo termine non nullo

b) Utilizzando questo risultato calcolare

$$\lim_{x\to 0} \frac{\sqrt{1+f(x)} - 1}{x^4}$$

$f(x) = \log(1+2x^2) + 4(\cos x - 1)$

$f(0) = 0 \qquad f'(x) = \frac{1}{1+2x^2} \cdot 4x - 4\sin x$

$f'(0) = 0$

$f''(x) = \frac{4(1+2x^2) - 16x^2}{(1+2x^2)^2} - 4\cos x$

$f''(0) = 0$

$f''(x) = \frac{4 - 8x^2}{(1+2x^2)^2} - 4\cos x$

$f'''(x) = \frac{-16x(1+2x^2)^2 - 2(1+2x^2)\cdot 4x(4-8x^2)}{(1+2x^2)^4}$

$+ 4\sin x \qquad f'''(0) = 0$

$f^{IV}(0) = -44$

$f(x) = \frac{-44}{4!}x^4 + o(x^4) = -\frac{11}{6}x^4 + o(x^4)$

$$\lim_{x\to 0} \frac{\sqrt{1 - \frac{11}{6}x^4} - 1}{x^4} = \lim_{x\to 0} \frac{\frac{1}{2}\left(-\frac{11}{6}x^4\right) + o(x^4)}{x^4}$$

$= -\frac{11}{12}$